

BRT Safety Filtering for LLM-Generated Orbital Maneuver Plans

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Abstract—Large Language Models (LLMs) are becoming considerably more appealing as a flexible interface/console for autonomous spacecraft mission planning due to their ability to turn language objectives into maneuver sequences. However, LLMs have no intrinsic understanding of orbital mechanics or physical safety constraints, and can generate plans that violate keep-out zones or lead to collision. We propose a safety filtering pipeline in which a precomputed Backward Reachable Tube (BRT), derived via Hamilton-Jacobi reachability over Clohessy-Wiltshire (CW) dynamics, intercepts unsafe LLM-generated commands and replaces them with minimum-norm safe alternatives. The 6D BRT has been computed and verified, and LLM plan generation and evaluation are the next steps.

Index Terms—Hamilton-Jacobi reachability, backward reachable tube, safety filtering, large language models, orbital proximity operations, Clohessy-Wiltshire equations.

I. INTRODUCTION AND MOTIVATION

AUTONOMOUS on-board mission planning is becoming essential for spacecraft that cannot rely on ground support. Deep-space missions can encounter communication delays of up to hours, and congested LEO orbits may require a spacecraft to handle potential collision events between contact windows. In both cases, the spacecraft must plan and execute maneuvers without waiting for human intervention.

LLMs have recently become an appealing component for such systems for their ability to turn language objectives into structured command sequences. An operator might specify things such as a specific approach angle and speed to a target, and an LLM could generate a corresponding maneuver sequence. The problem is that LLMs have no understanding of orbital mechanics or safety constraints, and thus may generate an unsafe plan that can drive a chaser straight into a prohibited region.

Reachability-based runtime monitors have been proposed for LLM-controlled robots [1], and formal methods have been applied to spacecraft planning using MDP frameworks [2], but no existing work applies Hamilton-Jacobi (HJ) reachability as a runtime safety filter for LLM-generated orbital maneuver plans. However, since LLMs can produce physically impossible or dangerous plans with no warning—and in proximity operations there is no recovering from a collision—formal safety layers are required. We propose a pipeline in which a precomputed BRT intercepts unsafe LLM commands at each timestep and replaces them with minimum-norm safe alternatives, providing a worst-case safety guarantee independent of the LLM’s internal reasoning.

II. RELATED WORK

A. Hamilton-Jacobi Reachability for Spacecraft Safety

Hamilton-Jacobi reachability computes the Backward Reachable Tube (BRT), defined as the set of states from which failure is inevitable regardless of control input [3], by solving an HJ partial differential equation (PDE) backward in time. This framework has been applied in various contexts to spacecraft proximity operations: Holzinger and Scheeres [4] derived reachability conditions for proximity operations and formation flight under Keplerian relative motion; Zagaris and Romano [5] characterized reachable docking sets for a rotating target; and de la Barcena [6] directly compares HJ reachability against control barrier function (CBF)-based approaches under CW dynamics, finding that HJ reachability is more suitable when hard guarantees and timely responses are required.

B. LLMs for Autonomous Mission Planning

Rodriguez-Fernandez et al. [7] investigated LLM-based satellite guidance, navigation, and control by engineering prompts on the KSPDG challenge, while Maranto [8] demonstrated that LLM planning performance degrades as mission complexity increases. Constrained LLM planning has been explored through temporal logic verification [9], [10] and SMT-based barrier synthesis [11]. Hafez et al. [1] propose reachability-based runtime monitors for LLM-controlled robots. However, none of these approaches enforce safety for continuous control inputs in orbital dynamical systems. Specifically, the planning and verification methods [9]–[11] function offline or on discrete action sequences, the spacecraft LLM work [7], [8] lack formal safety guarantees at the control level, and the reachability monitor of [1] is for general robot navigation rather than orbital proximity operations.

C. Alternative Safety Methods and Motivation for BRT Filtering

Breeden and Panagou [12], [13] apply Robust CBF quadratic programs to spacecraft docking, and Dunlap et al. [14], [15] show within the Run Time Assurance framework that safety filters can be decoupled from the primary controller, which is the same principle that we will apply. CBFs encounter two problems in our experimental setting. First, the position-based KOZ constraint has a relative degree of two under CW dynamics, as thrust affects velocity rather than position directly. Consequently, bounded inputs can make the QP infeasible near the boundary [13]. Secondly, impulsive ΔV burns

do not enforce constraints between timesteps. Other methods, such as stochastic reachability [16], [17] have been applied to spacecraft rendezvous and docking under CW dynamics to address uncertainty, but yield probability bounds instead of worst-case guarantees, and require approximation in 6D. HJ reachability addresses these limitations: the Hamiltonian is defined over any compact control set without smoothness assumptions, the BRT provides hard guarantees regardless of LLM output, and all computation is offline. The primary trade-off is that the BRT must be recomputed if the KOZ geometry changes, which is acceptable for our non-maneuvering, fixed-Local Vertical Local Horizontal (LVLH) frame KOZ scenario considered here.

III. PROBLEM FORMULATION

This work considers a chaser spacecraft approaching a non-maneuvering target in a low Earth orbit (LEO). We model relative motion using the linearized CW equations, assume perfect state knowledge, and fix the KOZ geometry for the duration of the approach. The goal is a runtime safety filter that blocks any LLM-commanded burn that would drive the chaser into the KOZ.

A. Relative Motion Dynamics

We model the chaser's motion relative to a target spacecraft in a near-circular orbit using the Clohessy-Wiltshire (CW) equations [18]. The state

$$x = [r_x, r_y, r_z, \dot{r}_x, \dot{r}_y, \dot{r}_z]^\top \quad (1)$$

gives relative position and velocity in the target-fixed LVLH frame. The linearized dynamics are $\dot{x} = Ax + Ba$, where

$$A = \begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 3n^2 & 0 & 0 & 0 & 2n & 0 \\ 0 & 0 & 0 & -2n & 0 & 0 \\ 0 & 0 & -n^2 & 0 & 0 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 0_{3 \times 3} \\ I_{3 \times 3} \end{bmatrix}, \quad (2)$$

n is mean motion, and $a \in \mathcal{U}$ is bounded thrust acceleration with $a_{\max} = 0.2 \text{ m/s}^2$. BRT computation uses continuous, bounded acceleration, whereas simulation uses impulsive ΔV burns, which are exact for the linearized model.

B. Keep-Out Zone

The target is surrounded by an ellipsoidal *failure set*

$$\mathcal{F} = \{x : (r - c)^\top E (r - c) \leq 1\}, \quad (3)$$

where $r = [r_x, r_y, r_z]^\top$, c is the KOZ center, and $E = \text{diag}(a^{-2}, b^{-2}, c_z^{-2})$ encodes semi-axes (a, b, c_z) .

C. Backward Reachable Tube

The BRT $\mathcal{B}(t)$ is the set of states from which entry into $\mathcal{F}$ is unavoidable over $[t, T]$ regardless of control:

$$\mathcal{B}(t) = \{x_0 : \forall u(\cdot) \in \mathcal{U}, \exists \tau \in [t, T] \text{ s.t. } x(\tau) \in \mathcal{F}\}. \quad (4)$$

It is characterized by the solution V of the HJ PDE [19]

$$\frac{\partial V}{\partial t} + \min(0, H(x, \nabla_x V)) = 0, \quad V(x, T) = l(x), \quad (5)$$

where $l(x) = (r - c)^\top E (r - c) - 1$ (negative inside $\mathcal{F}$) and

$$H(x, p) = \min_{a \in \mathcal{U}} p^\top (Ax + Ba), \quad p \triangleq \nabla_x V \in \mathbb{R}^6. \quad (6)$$

The BRT and safe set follow directly from the sign of V : $\mathcal{B}(t) = \{x : V(x, t) \leq 0\}$, $\mathcal{B}^c(t) = \{x : V(x, t) > 0\}$.

D. LLM Maneuver Plan

Let $\pi_{\text{LLM}} = \{(k, \Delta V_k)\}_{k=0}^N$ denote a plan produced by the LLM, where $\Delta V_k \in \mathbb{R}^3$ is the commanded velocity increment at discrete timestep k .

E. Safety Filtering Problem

At each timestep k , the filter first checks whether the LLM command is already safe by evaluating $V(f(x_k, \Delta V_k), t_{k+1})$. If this value is positive, x_{k+1} lies in the safe set and the command simply passes through: $\Delta V_k^* = \Delta V_k$. If it is non-positive, x_{k+1} would enter the BRT and the filter solves

$$\Delta V_k^* = \arg \min_{\Delta V} \|\Delta V - \Delta V_k\| \quad \text{s.t.} \quad V(f(x_k, \Delta V), t_{k+1}) > 0, \quad (7)$$

to instead return the minimum-norm correction by finding the largest safe scaling of the original burn: $\Delta V_k^* = \alpha \Delta V_k$, $\alpha \in [0, 1]$.

IV. PROPOSED APPROACH

A. BRT Precomputation

The BRT is computed offline using the `hj_reachability` JAX-based solver [20], propagating the HJ PDE backward over a 300s horizon on an $11 \times 35 \times 11 \times 8 \times 8 \times 8$ grid spanning $[\pm 1200, \pm 600, \pm 600]$ m in position and ± 3 m/s in each velocity axis. The value function is stored as a compressed `.npz` array and queried at runtime with a `scipy RegularGridInterpolator`.

B. LLM Prompting Pipeline

We use an LLM (Anthropic API) to generate proximity maneuver plans from natural language, with each prompt providing the scenario geometry, the current chaser state, and a mission specification $\mathcal{M}$ to request a structured JSON output, e.g.:

```
{
  "maneuvers": [
    {"t_s": 45, "dv_m_s": [-0.02, 0.1, 0.0],
     "kind": "approach"},
    {"t_s": 60, "dv_m_s": [0, 0, 0],
     "kind": "coast"}
  ]
}
```

A collection of 50–100 plans will be produced across varying mission specifications such as approach angle, urgency, and initial range to the KOZ boundary.

C. Runtime Safety Filter

At each timestep k , the filter queries $V(x_{k+1}^{\text{LLM}}, t_{k+1})$. If $V \leq 0$, it runs a line-search from ΔV_k toward zero, accepting the largest scaled value with $V > 0$. An additional passive safety check verifies that free drift from x_k stays outside $\mathcal{F}$ over a finite horizon, providing an additional safety margin independent of future burns.

D. Evaluation and Baselines

Three conditions will be evaluated over the collection of plans: (1) no filter, (2) BRT filter, and (3) a rule-based planner using fixed radial burns as a baseline. The primary metrics are interception rate, mean ΔV correction overhead, and mission success rate (chaser reaches target without entering $\mathcal{F}$), reported before and after filtering.

V. PRELIMINARY RESULTS

The 6D BRT has been computed for a 400 km circular LEO orbit, with KOZ semi-axes $[28, 45, 18]$ m, $a_{\text{max}} = 0.2 \text{ m/s}^2$, and a 300 s backward horizon. Figs. 1 and 2 show the x - y slice of V at $z = 0$, $v \approx 0$ at $\tau = -164$ s and $\tau = -300$ s, confirming the BRT grows as $|\tau|$ increases. Fig. 3 shows the chaser orbit in ECI coordinates alongside the LVLH formation view with the calculated BRT encompassing the target.

The pipeline was validated on a three-burn approach sequence (initial along-track separation 3200 m, $\Delta V \sim 10^{-2}$ m/s per burn). At each segment boundary, the BRT value function is queried and the passive drift check is run. Table I confirms that all states are both outside the BRT and passively safe, as expected at a 3 km range. The LLM JSON parser is also complete and has been tested on manually constructed plans.

TABLE I
BRT EVALUATION: 3-BURN APPROACH SEQUENCE

Time (s)	$V(x)$	BRT unsafe?	Passively safe?
0	5353	No	Yes
45	5325	No	Yes
90	5290	No	Yes
150	5254	No	Yes

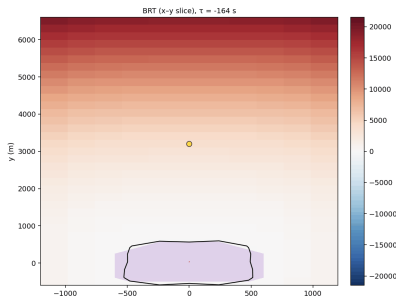


Fig. 1. x - y slice of the HJ value function V ($z = 0$, $v \approx 0$) at $\tau = -164$ s. The black contour marks $V = 0$ and the shaded region is the BRT ($V \leq 0$). The orange marker shows the chaser at 3200 m, inside the safe set.

This verifies that the current pipeline components are operating as intended. However, the BRT calculation process is still being finalized. We are currently exploring various

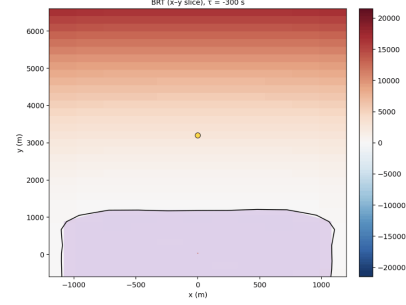


Fig. 2. x - y slice at $\tau = -300$ s. The BRT has grown relative to Fig. 1, confirming correct backward-reachability semantics.

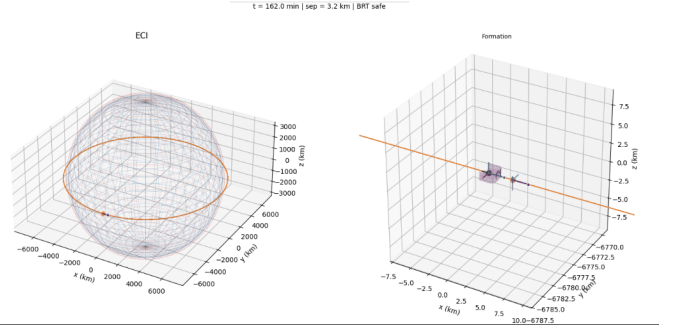


Fig. 3. Left: chaser orbit (orange) and Earth in ECI coordinates at $t = 162$ min. Right: LVLH formation view showing the chaser trajectory relative to the target (chief), with the BRT isosurface. Separation is 3.2 km; the state is BRT safe.

grid resolutions, domain sizes, and time horizons to assess how each affects the shape and numerical accuracy of the value function. Once a stable configuration is found, a higher-resolution computation will generate the final BRT for evaluation. For project success, we expect to achieve quantitative interception rate, mean ΔV correction overhead, and mission success rate metrics for our collection of LLM-generated plans. The immediate prerequisites are LLM API integration and testing BRT entry detection scenarios in the current pipeline.

VI. MILESTONES AND WORK PLAN

The milestones for Weeks 6–7 have been completed. The CW dynamics model, preliminary 6D BRT calculation, and ellipsoidal KOZ have been implemented, and the safety evaluation pipeline, including BRT querying, passive drift check, trajectory logging, is operational. The tasks for Week 8 are currently in progress, including the prompting pipeline, minimum-norm correction implementation, and full plan generation. Weeks 9 and 10 will focus on metric evaluation, baseline comparisons, potential edge-case testing, and writing of the final report.

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